CURRICULUM VITAE

Ali A. Alhejab.

Lead AI Engineer · HUMAIN · Riyadh

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PROFILE

Lead AI Engineer with hands-on experience across agentic AI, LLMs, generative image and video, deep learning, and computer vision. I build production systems end to end — from real-time conversational agents and custom avatar platforms to diffusion model training, inference optimization, and multi-GPU pipelines.

PROFESSIONAL EXPERIENCE

Lead AI Engineer

HUMAIN · Riyadh, Saudi Arabia

May 2025 – Present

- Built an in-house avatar platform, achieving a 92% cost reduction versus third-party solutions.
- Developed and deployed real-time AI avatar agents as interactive virtual presenters and meeting facilitators, with conversational pipelines integrating STT, LLM, and TTS.
- Pretrained and fine-tuned video diffusion models for custom avatar generation, optimizing inference for both CPU and GPU.
- Built an in-house intent-detection model achieving 99% accuracy to route user requests within the conversational pipeline.

AI Engineer

SDAIA · Saudi Arabia

Jun 2024 – May 2025

Research and development of AI models across computer vision, NLP, and generative modeling.

- Trained DETR and YOLO object-detection models for construction-site surveillance (helmets, vehicles, license plates, and people).
- Built a sign-language-to-text translation system using T5 and AV-HuBERT; contributed to the Sign4All Challenge.
- Fine-tuned LoRA modules for diffusion models (Flux, Hunyuan, Wan 2.1) for image and video generation.
- Fine-tuned LLaMA and T5 for token-by-token image generation/understanding and conversational abilities.
- Optimized inference with TensorRT (vision) and vLLM (LLMs); proficient with TeaCache for generation workflows.
- Built scalable multi-GPU / multi-node training pipelines using HuggingFace Accelerate and DeepSpeed.

Solar Energy Systems Researcher

KAUST · Saudi Arabia

Aug 2022 – May 2024

Developed techniques to localize and identify faults in PV systems.

- Collaborated with the KAUST Solar Center to build a fast, cost-effective fault-localization algorithm for PV systems.
- Partnered with NEOM to test new solar-glass technologies.

Technical Services Division Trainee

Ithra, Saudi Aramco · Saudi Arabia

Jun 2021 – Aug 2021

- Member of the quality and maintenance inspection team during a summer internship.

SELECTED PUBLICATIONS

01 SPRINGER 2026

Saudi Sign Language Translation Using T5

A. Alhejab et al.

Springer, Cham

02 IEEE 2025

A Single-Voltage-Sensor Bypass-Switch-Based Photovoltaic Fault Localization

A. Alhejab, M. Abbasi, S. Ahmed

IEEE Journal of Photovoltaics

EDUCATION

King Abdullah University of Science and Technology (KAUST)

Jun 2024

Thuwal, Saudi Arabia

M.Sc., Electrical and Computer Engineering

Specialization in Solar Energy Systems · GPA: 3.67/4.0

Thesis: "A Novel Fault Localization Technique for PV Systems Using a Single-Voltage Sensor." Advisor: Prof. Shehab Ahmed.

King Fahd University of Petroleum and Minerals (KFUPM)

May 2022

Dhahran, Saudi Arabia

B.Sc., Electrical Engineering

Specialization in Internet of Things · GPA: 3.75/4.0

Research: "Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter." Advisor: Dr. Yaqub Mahnashi.

TECHNICAL SKILLS

PROGRAMMING

Python, C/C++, MATLAB

SOFTWARE & TOOLS

Hugging Face, LLaMA-Factory, PyTorch, Git, Jupyter, KiCad, LTspice, LaTeX

RELEVANT COURSEWORK

Deep Learning for Visual Computing, Mathematics for Machine Learning, Renewables, Microwave Circuits, Computer Networks, IoT, Energy Harvesting

TEACHING

Deep Learning and Reinforcement Learning

LANGUAGES

Arabic (native), English (advanced)

03 IEEE 2024

Energy-Efficient Aerial Data Aggregation for IoT: From Prototyping to Large-Scale Implementation
O. Khalifa, A. S. Mohammed, A. Alhejab, et al.
IEEE Transactions on Instrumentation and Measurement

04 IEEE 2023

Energy-Efficient Aerial Data Aggregation in IoT Networks with WuR
A. S. Mohammed, O. Khalifa, A. Alhejab, et al.
IEEE International Black Sea Conference on Communications and Networking

05 AUTHOREA 2024

A New Synthesis Method for Efficient Reconfigurable Switched-Capacitor Converters
A. Alhejab, Y. Mahnashi
Authorea Preprints

CERTIFICATIONS

NVIDIA Certified Associate — Generative AI (Multimodal) & Generative AI (LLM)
Riyadh, Saudi Arabia · Dec 2024

PV Design & Installation School
Thuwal, Saudi Arabia · Apr 2023 · 80-hour program on PV system design, installation, and commissioning

Self-Driving Car Engineer Nanodegree
Online · Jun–Oct 2021 · Misk-sponsored, project-based program covering control systems, machine learning, and sensor fusion

HONORS & AWARDS

1st Place — Abu Dhabi University 11th Research Competition, 2023.
Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter.

1st Place — KFUPM Senior Design Project EXPO, 2022.
Low-Power IoT Data Aggregation System Using a UAV as Data Aggregator.